

BARTOSZ STARZYK

SOFTWARE & EMBEDDED DEVELOPER

Detail-oriented Software Engineer specializing in embedded systems and low-level programming (C++, Rust). Proven commercial experience in developing Digital Twins for the energy sector and medical applications within Industry 4.0 standards. Skilled in data optimization and E2E test automation.

EXPERIENCE



Software & QA Intern March – April 2025
Digital Technology Poland (DTP)

- Development of Digital Twin software for the energy sector using E-CAD/CAD systems
- Automated E2E tests using Playwright, significantly improving system stability and regression verification
- Optimized data workflows for Industry 4.0 projects in high-reliability environments
- Worked according to strict industrial standards and commercial processes

Mobile App Developer Intern May – June 2024
Healthcare Center (ZZOZ) in Cieszyn

- Designed and deployed a Kotlin-based mobile application for digital medical equipment inventory
- Managed the full development lifecycle: from requirement analysis (UX/UI) to production deployment
- Collaborated closely with medical staff to tailor the tool to daily operational needs
- Ensured data security compliance with Healthcare IT standards

FEATURED PROJECTS



3D Model Inspector [C++ / OpenGL]

A high-performance tool for importing and inspecting OBJ models with PBR (Physically Based Rendering) support and directional lighting.

Raspberry Pico Console [C / Embedded / RP2040]

A DIY handheld gaming console featuring a modular game loader and custom PCB. Implemented games including a 2.5D maze renderer.

Lucario Engine [Rust / Vulkan]

Procedural voxel world engine in Rust with a custom Vulkan renderer, greedy meshing and multi-threaded chunk generation.

[lshanterl.github.io](https://github.com/lshanterl)
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 Full Category B



SKILLS

Programming Languages

- **Main:** C++, Rust, Python
- **Additional:** Kotlin, C#, Typescript, Go

Embedded & Hardware

- **Platforms:** RP2040, ESP32, ARM
- **Interfaces:** I2C, SPI, UART, GPIO
- **Design & Hardware:** PCB prototyping, soldering & measurements, 3D modeling

Tools & Technologies

- **Systems:** Git, CMake
- **QA & Dev:** Playwright (E2E), CI/CD (podstawy)
- **Graphics:** Blender, OpenGL, Unity, Godot, Fusion 360

Professional Competencies

- Requirements Analysis & System Design (UX/UI)
- Teamwork (Agile/Scrum)
- Technical communication (collaboration with clients/staff)
- Rapid adaptation to new technologies

EDUCATION

International Training

Schkeuditz – Erasmus +

- Intercultural communication
- Working in an English-speaking environment

Wł. Szybiński Technical Secondary School, Cieszyn 2021 – 2026

Technician (Programmer)

LANGUAGES

- Polish – native
- English – advanced

I hereby give consent for my personal data included in my application to be processed for the purposes of the recruitment process under the Personal Data Protection Act as of 10 May 2018 (Journal of Laws 2018, item 1000) and the Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 (GDPR).